

JIS COLLEGE OF ENGINEERING

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MAJOR PROJECT - I

REVIEW 1

Blockchain-Assisted Distributed AES Key Lifecycle Management Framework for Secure Healthcare Internet of Things

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1. Background

Healthcare Internet of Things (IoT) ecosystems continuously generate and exchange highly sensitive medical information among wearable devices, sensors, cloud servers, physicians, and healthcare institutions. AES provides efficient encryption for protecting patient data; however, centralized key management can introduce single points of failure, insider threats, unauthorized modifications, and poor auditability.

Blockchain provides decentralized trust, immutability, and transparent transaction logging, making it suitable for distributed cryptographic key management.

2. Problem Statement

The project aims to develop a blockchain-based distributed AES key lifecycle management framework for healthcare IoT ecosystems. The framework will securely manage cryptographic keys throughout their lifecycle, including generation, distribution, storage, periodic rotation, revocation, archival, and emergency recovery through blockchain-enabled smart contracts.

Patient records will remain encrypted using AES-256. Blockchain will maintain cryptographic metadata, audit logs, authorization policies, and key lifecycle events without exposing confidential medical information.

3. Existing Approach and Limitations

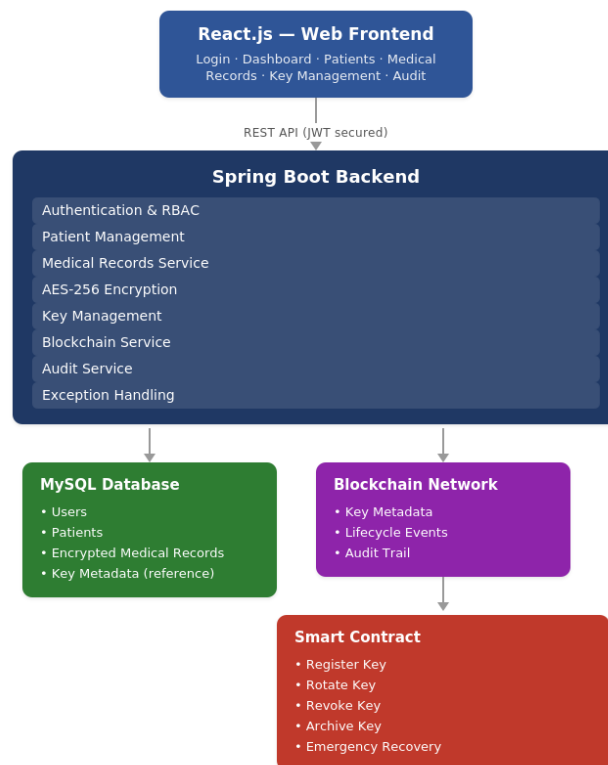
Existing Approach	Major Limitations
AES encryption with centralized cryptographic key management.	Single point of failure; insider threats; unauthorized modifications; limited auditability and traceability of key operations.
Manual / administrator-driven key lifecycle handling (generation, rotation, revocation).	Error-prone, difficult to scale, and lacks a tamper-resistant history of who performed which key operation and when.
Centralized logging for compliance and audit.	Logs can be altered or deleted by a privileged insider, weakening accountability and regulatory compliance.

4. Proposed Solution and Architecture

The proposed system combines AES-256 encryption with blockchain-assisted distributed key lifecycle management. Medical records are encrypted and stored outside the blockchain (in a relational database), while the blockchain records only cryptographic metadata, authorization policies, audit information, and key lifecycle events. This ensures that no confidential medical information is ever exposed on-chain, while still gaining blockchain's tamper-resistance and traceability for key governance.

In simple terms: the framework protects sensitive healthcare data using AES-256, and separately manages the complete lifecycle of the encryption keys using blockchain and smart contracts — providing controlled key operations, traceability, and tamper-resistant auditing, without ever storing confidential medical data on the blockchain.

4.1 System Architecture



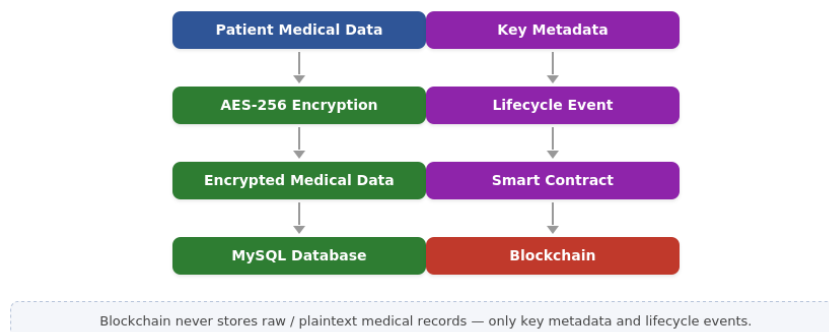
4.2 User Roles

The system defines four primary user roles, each with distinct access privileges enforced through authentication and authorization (JWT-based RBAC):

- Admin — manages users, system configuration, and oversees overall key governance.
- Doctor — accesses authorized patients' medical records for diagnosis and treatment.
- Patient — views their own medical records and access history.
- Auditor — inspects key lifecycle events and audit trails for compliance review.

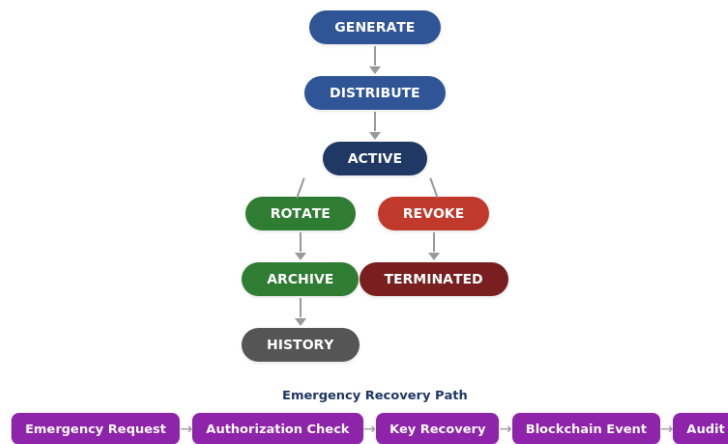
4.3 Medical Data Flow

Medical data and cryptographic metadata follow two clearly separated paths — only the encrypted data touches the database, and only non-sensitive key metadata touches the blockchain:



4.4 Key Lifecycle Management

The AES-256 key lifecycle is governed end-to-end through blockchain-enabled smart contracts, covering normal operation as well as emergency scenarios:



4.5 Initial Database Design

A minimal relational schema is used to store users, patients, encrypted medical records, and key metadata; blockchain-related tables (transaction hash references) link the two layers together.

Table	Key Fields
users	id, name, email, password, role, created_at
patients	id, user_id, patient_code, date_of_birth, created_at
medical_records	id, patient_id, encrypted_data, key_id, key_version, created_at, updated_at
key_metadata	id, key_identifier, version, status, algorithm, created_at, rotated_at, revoked_at
key_lifecycle_events	id, key_id, operation, performed_by, timestamp, status, blockchain_tx_hash

4.6 Backend and Frontend Structure

Backend (Spring Boot) follows a layered, service-oriented package structure; Frontend (React.js) mirrors it with a dedicated page for each core function:



4.7 Blockchain Responsibility (Scope Boundary)

Blockchain DOES store • Key metadata references • Key lifecycle events • Authorization-related records • Audit trail • Tamper-resistant history	Blockchain does NOT store • Raw / plaintext medical records • Plaintext passwords • Sensitive confidential information • Raw AES secret keys
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5. Research Objectives

- Eliminate centralized key management.
- Improve cryptographic transparency and accountability.
- Automate secure key lifecycle operations.
- Evaluate blockchain overhead.
- Improve resilience against insider attacks.

6. Major Modules / Tangible Deliverables

- Healthcare Encryption Platform
- Blockchain Network
- Smart Contract-based Key Management System
- AES Encryption Module
- Secure Key Rotation Engine
- Audit and Compliance Dashboard
- Security Evaluation Framework

7. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Java + Spring Boot
Database	MySQL
Security / Encryption	Spring Security, JWT, AES-256
Blockchain	Private / Local Blockchain + Smart Contracts
Development & Testing	Maven, Git/GitHub, Postman

8. Expected Outcome

A working prototype that securely encrypts healthcare records using AES-256 and provides distributed, traceable, and auditable cryptographic key lifecycle management using blockchain and smart contracts. The project will also evaluate security and blockchain-related performance overhead.

9. Review 1 Development Plan

- Finalize system requirements and architecture.
- Implement authentication, authorization, healthcare data model, and basic REST APIs.
- Implement and test the initial AES-256 encryption module.
- Design the key metadata and lifecycle model.
- Prepare the initial blockchain and smart-contract integration plan.